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3 Hours / 70 Marks Seat No. | | | | | | | | |

Instructions : (1) All Questions are *compulsory.*

(2) Answer each next main Question on a new page.

(3) Illustrate your answers with neat sketches wherever necessary.

(4) Figures to the right indicate full marks.

(5) Assume suitable data, if necessary.

Marks

1. Attempt any FIVE of the following : 10

(a) Draw the flow chart of conversion of yarn to fabric.

(b) List down types of loom used for woven fabric production.

(c) Define following term :

(1) Metric yarn count

(2) Denier

(d) Explain with neat diagram different types of yarn faults (any two).

(e) Classify different types of winding machine.

(f) List down different yarn joining methods.

(g) Define following term :

(1) Traverse length

(2) Traverse ratio

2. Attempt any THREE of the following : 12

(a) Compare direct and indirect yarn numbering systems.

(b) Describe classimat yarn faults.

(c) Compare drum winding with precision winding machine.

(d) List down and explain any two winding package faults with neat diagram.

3. Attempt any THREE of the following : 12

(a) Find the count of 34 grams of 7.2 km yarn in English count.

(b) State the significance of objectionable faults in the winding.

(c) Explain Gate type tentioner with neat diagram.

(d) Explain :

(1) Cleaning efficiency

(2) Knot factor

4. Attempt any THREE of the following : 12

(a) Describe flow chart of woven fabric production with objectives of each process.

(b) Calculate length in metres of 60^S cotton yarn wound on winding cone of weight 2.5 kg.

(c) Explain with neat diagram different end packages in winding.

(d) Explain electronic yarn clearing devices with neat diagram.

(e) Explain the term :

(1) Coil angle

(2) Winding speed

5. Attempt any TWO of the following : 12

(a) Describe passage of yarn through core winding machine with neat sketch.

(b) State the role of tention in winding. Also state factors affecting yarn tention during winding.

(c) Calculate the production of core winding machine with following particulars :

Drum speed – 900 rpm

Drum dia. – 3 inches

No. of drum – 60

Shift hours – 8 hours

Efficiency – 75%,

Count of yarn – 40^S Ne

6. Attempt any TWO of the following : 12

(a) Select the finer yarn from the wooden box containing 3 bobbins-100 Denier polyester, 60^S Ne cotton yarn, 40 Tex polyester yarn with justification

(b) Explain pneumatic splicing with neat diagram. Also state the advantages of it.

(c) Yarn are required to wind 1600 lb of 50^S Ne cotton count in 8 hours on high speed winding machine with speed 650 YPM. Machine runs with 89% efficiency. Calculate how many drums were required.